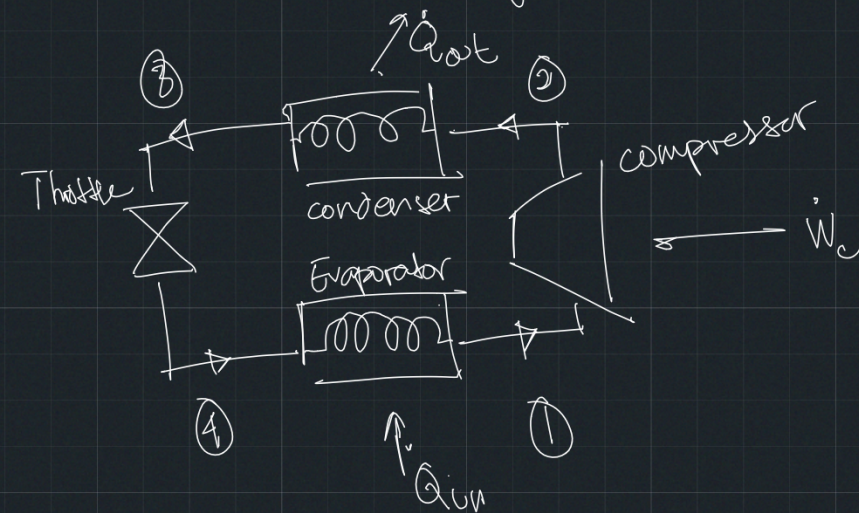


9/25 Vapor compression system

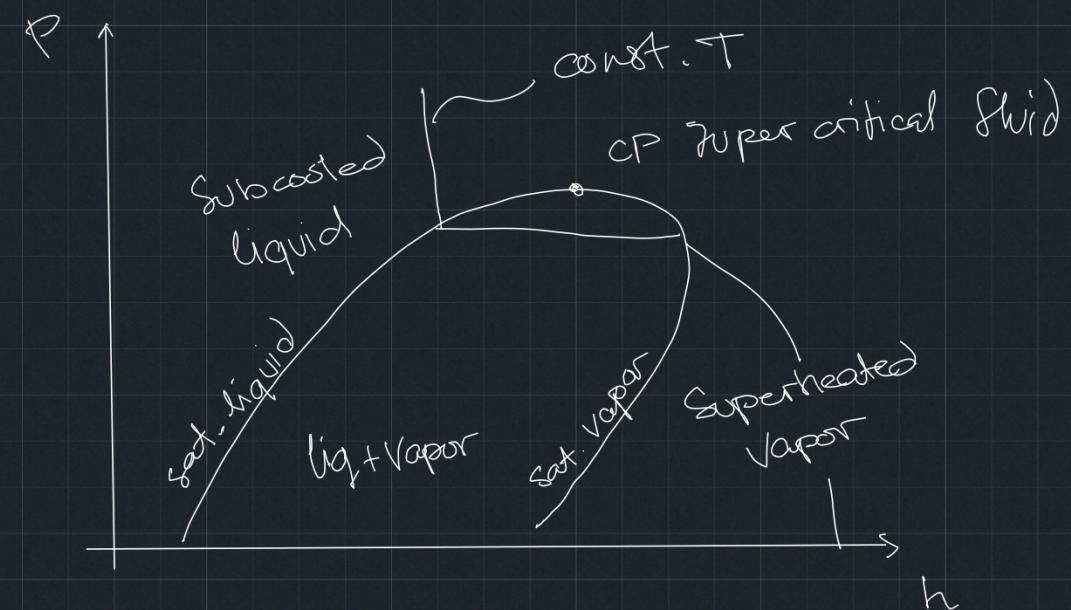


evaporator $\dot{Q}_{in} = \dot{m}(h_1 - h_4)$

condenser $\dot{Q}_{out} = \dot{m}(h_2 - h_3)$

throttle $h_3 = h_4$

compressor $\dot{W}_c = \dot{m}(h_2 - h_1) + \dot{Q}_c$



Psychrometrics



$$\phi = \frac{P_v}{P_{sat}}$$

CH Conservation laws

Mass: $\frac{dM_{tot}}{dt} = \sum \dot{m}_{in} - \sum \dot{m}_{out}$

Energy: $\frac{dE_{tot}}{dt} = \sum \dot{Q}_{in} - \sum \dot{W}_{out} + \sum \dot{W}_{in} - \sum \dot{Q}_{out}$
 $+ \sum \dot{m}_{in} \left(h_{in} + \frac{1}{2} V^2 + g z \right)_{in}$
 $+ \sum \dot{m}_{out} \left(h + \frac{1}{2} V^2 + g z \right)_{out}$